



The Clovis culture

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Death of the megabeasts

What really killed the North American Megafauna remains a mystery.
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According to the theory, the explosion or impact by one or more cometary fragments, destabilised the ice sheets, and was a contributing factor in the extinction of the American megafauna, as well as the sociocultural changes in the societies of the people at that time. The heat from the impact melted the ice sheets, pumping fresh-water into the oceans, throwing the warming currents out of whack, and leading to a global cooldown.

Since the original presentation of the paper in 2007, there have been a number of formal debates organised between proponents and opponents of the hypothesis. The study was immediately controversial as it could not conclusively prove that the markers that it had identified as signs of the impact event, including nanodiamonds, carbon spheres, iridium and radioactivity, which can all be considered evidence of an extra terrestrial object entering the Earth's atmosphere, or even an impactor. Subsequent studies could not corroborate the findings of the original, and a number of terrestrial processes have been identified that could have given rise to the identified markers. The presence of the charred carbon rich layer can be attributed to a series of fires, instead of one cataclysmic continent spanning inferno. The forest fires that raged across the continent may have themselves given rise to some of the observed indicators of a potential impact, such as the nanodiamonds. There is also another potential identified cause of the fires instead of an impactor - humans themselves. The prevailing theory is the is that runoff from melting glaciers as the world warmed disrupted the oceanic currents, temporarily halting the world from heating up. Studies of Antarctic ice sheets and ancient Chinese stalagmites have indicated that there might have been other cooling events similar to the Younger Dryas, during other interglacial periods on the planet, when the Earth was moving from an icehouse planet to a greenhouse

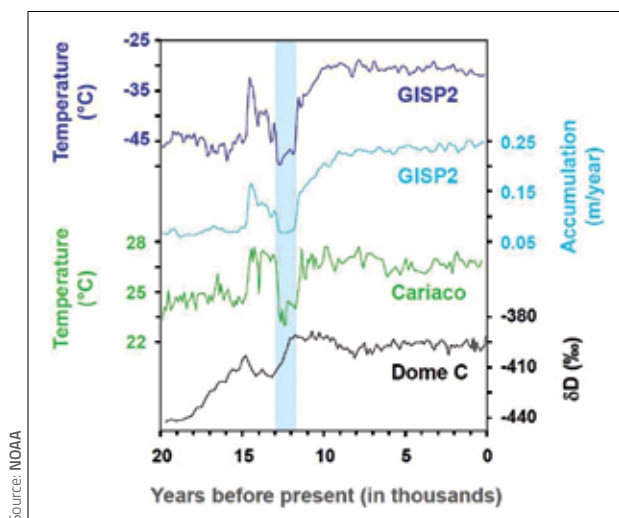
one. The point in contention here is not what caused the Younger Dryas cooling, but whether or not one or multiple impactors also contributed to more freshwater landing up in the oceans, and if one of these causes was an extraterrestrial impactor.

When it comes to the extinction of the megafauna, there are contrasting ways of reading the data. A number of different proxies and markers are used to date the extinction of the megafauna, and there is some evidence to suggest that the extinction in North America happened before the onset of the Younger Dryas, that is before any supposed impactor. Here the

tools and kill sites did not have bones of some animals that did go extinct, such as the giant sloth. When it comes to the question of what exactly led to the extinction of the American megafauna, scientists do not have a definite answer.

Then there is the question of whether or not there was a population decline around the time of the Younger Dryas. There is evidence available both ways, but the opponents to the Younger Dryas impact hypothesis fortify their argument by pointing out certain sites where the Clovis people seemed to overlap with that of other, later cultures. In Syria,

there is evidence that human populations survived without being disturbed even if an airburst of an extraterrestrial impactor happened nearby. The problem with the event is that it left behind little traces if it happened, leaving researchers trying to theoretically approach a problem that has many possible explanations. Even if the fragments of the comet struck



Source: NOAA

The blue bar shows a drop in temperatures in Greenland and near Venezuela, with the black graph representing temperatures in Antarctica, which rose during the Younger Dryas.

science gets murkier, as neither the dates nor the causes of the extinction of the megafauna are widely agreed upon. Possible explanations for the extinction include raging pandemics, ecosystem changes, extinction of keystone species and a "blitzkrieg" like overkill from humans who had just crossed the Bering strait into a world full of... well food. Overkill is the most favoured explanation, but there are certain discrepancies in that as well. Some species, such as Bison, Elk and Moose increased in number during the human occupation, before the extinction of wild horses and mammoths. At the same time Clovis

the planet, they did so on top of ice sheets, leaving behind no specific craters that can be linked to the Younger Dryas impact.

The Younger Dryas impact hypothesis is heavily criticized for reasons that have mostly to do with how the available data is interpreted. For this reason, there is new evidence unearthed all the time that either seems to support or contradict the hypothesis. The theory keeps getting "proved" and "disproved". While the scientists do not know whether there was an impact at all, neither do they fully understand the causes and timings of the Younger Dryas. ❗